

Comprehensive Blueprint: Modern Privacy-Enhancing Technologies (PETs)

Risk Assessment & Data Privacy Engineering

Contents

1 Introduction

This document outlines the full landscape of modern privacy enhancement techniques, providing the formal mathematical definitions, step-by-step algorithmic logic, and a clear mapping of which vulnerabilities they neutralize. The techniques are grouped into five primary families.

2 Family 1: Statistical Disclosure Control (SDC)

SDC techniques alter the data directly to minimize the risk of identifying individuals while preserving the statistical utility of the dataset.

2.1 Suppression & Generalization (k -Anonymity)

Logic: Hides individuals in a crowd by ensuring that any individual's quasi-identifiers (QIs) are identical to at least $k - 1$ other people in the dataset.

Math: Let T be a table and QI_T be the set of quasi-identifiers. T satisfies k -anonymity if for every tuple $t \in T$, there exists a set of tuples $E \subseteq T$ (an equivalence class) such that $|E| \geq k$ and $\forall t_i \in E, t_i[QI_T] = t[QI_T]$.

Algorithm (Mondrian Greedy Partitioning):

1. Initialize a single partition containing all data points.
2. For each partition, choose the QI attribute with the widest normalized scale range.
3. Find the median split point for that attribute.
4. If splitting results in two sub-partitions where both contain $\geq k$ records, execute the split.
5. Repeat recursively until no further valid splits can be performed.
6. **Generalization:** Replace exact values within each final partition with the bounding range $[min, max]$.
7. **Suppression:** If any leftover individual records form a group smaller than k , delete those rows entirely.

2.2 l -Diversity

Logic: Extends k -anonymity. l -diversity forces each equivalence class to contain at least l "well-represented" sensitive values to prevent attribute disclosure when a k -group shares the same sensitive value.

Math (Entropy-Based): For each equivalence class E , let S be the sensitive attribute, and $p(s)$ be the fraction of records in E that take the value $s \in S$. The class E satisfies entropy l -diversity if:

$$-\sum_{s \in S} p(s) \log p(s) \geq \log(l)$$

Algorithm:

1. Generate equivalence classes using a k -anonymity partitioner.
2. For each partition E , compute the empirical distribution $p(s)$ of the sensitive attribute.
3. Calculate the Shannon entropy of $p(s)$.
4. If entropy $< \log(l)$, merge E with its physically nearest neighboring partition.
5. Repeat the verification check until all partitions satisfy the constraint.

2.3 t -Closeness

Logic: Forces the distribution of sensitive attributes in any local group to be mathematically close to the global distribution, preventing inference attacks based on local skewness.

Math: Let P be the probability distribution of a sensitive attribute in an equivalence class E , and Q be the global distribution in the entire dataset. E satisfies t -closeness if the Earth Mover's Distance (EMD) $D[P, Q]$ satisfies:

$$D[P, Q] \leq t$$

Algorithm:

1. Compute the global frequency vector Q for all sensitive attributes.
2. Partition the dataset into k -anonymous groups.
3. For each partition, calculate its local frequency vector P .
4. Compute the EMD by solving the transportation linear program.
5. If $D[P, Q] > t$, iteratively merge the partition with adjacent partitions until the distance falls below t .

2.4 Data Swapping & Rank Swapping

Logic: Breaks the linkage between quasi-identifiers and sensitive values by swapping values between pairs of records while preserving individual univariable statistics.

Math: Let r_i, r_j be two records. In rank swapping, records are sorted by attribute A , and swapping is restricted to records whose rank index difference satisfies $|\text{rank}(r_i) - \text{rank}(r_j)| \leq p$.

2.5 Microaggregation

Logic: Replaces precise values with group averages to preserve aggregate statistical patterns while eliminating specific distinct records.

Math: For a cluster $C = \{x_1, x_2, \dots, x_k\}$, the vectors are replaced by the centroid vector:

$$\bar{x} = \frac{1}{k} \sum_{i=1}^k x_i$$

2.6 PRAM (Post Randomisation Method)

Logic: Probabilistic perturbation technique for categorical data using a known transition matrix.

2.7 Top/Bottom Coding & Noise Addition

Logic: Top/Bottom coding caps outliers at fixed thresholds. Noise addition injects continuous random values directly onto individual numeric entries.

3 Family 2: Differential Privacy (DP)

DP provides mathematical proof that an individual's presence or absence in a dataset will not significantly alter the output of an algorithm.

3.1 Global vs. Local DP

In Global DP, noise is added by a trusted curator. In Local DP, users add noise to their own answers before sending them.

3.2 The Laplace Mechanism

Logic: Adds noise drawn from a Laplace distribution to a numeric query output, calibrated to the query's global sensitivity.

Math: A mechanism $\mathcal{M}(D) = f(D) + Y$ provides ϵ -differential privacy if $Y \sim \text{Laplace}(0, \Delta f / \epsilon)$, where the global sensitivity Δf is defined as:

$$\Delta f = \max_{D_1, D_2} \|f(D_1) - f(D_2)\|_1$$

The probability density function of the noise is:

$$h(y) = \frac{\epsilon}{2\Delta f} \exp\left(-\frac{\epsilon|y|}{\Delta f}\right)$$

3.3 The Gaussian Mechanism

Logic: Adds Gaussian noise, yielding a relaxed (ϵ, δ) -DP guarantee.

$$\sigma \geq \frac{\Delta_2 f \sqrt{2 \ln(1.25/\delta)}}{\epsilon}$$

3.4 The Exponential Mechanism

Logic: Used for categorical outputs. It samples from all possible answers with a probability proportional to:

$$\exp\left(\frac{\epsilon \cdot u(D, r)}{2\Delta u}\right)$$

4 Family 3: Synthetic Data Generation (SDG)

SDG builds mathematical models of a dataset to generate completely artificial records.

4.1 Differentially Private SDG (DP-CGAN / DP-CTGAN)

Logic: Injects Differential Privacy directly into the training loop of a generative model.

Math (DP-SGD): During optimization, gradients $g_t(x_i)$ are clipped using threshold C and combined with Gaussian noise:

$$\tilde{g}_t = \frac{1}{B} \left(\sum_{i=1}^B \frac{g_t(x_i)}{\max\left(1, \frac{\|g_t(x_i)\|_2}{C}\right)} + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \right)$$

5 Family 4: Cryptographic Privacy-Enhancing Technologies

5.1 Homomorphic Encryption (HE)

Logic: Allows mathematical computations to be performed directly on encrypted values (ciphertexts).

Math (Paillier Cryptosystem - partially homomorphic for addition):

$$E(m_1) \cdot E(m_2) \equiv E(m_1 + m_2) \pmod{n^2}$$

5.2 Secure Multi-Party Computation (SMPC)

Logic: Splits data into encrypted, random fragments across multiple independent servers.

6 Family 5: Federated and Distributed Learning

6.1 Federated Learning with Secure Aggregation

Logic: Keeps training data decentralized. Nodes train locally and share only model weight adjustments.

Math (Federated Averaging - FedAvg):

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_t^k$$

7 Matrix: Attack Mitigation Mapping

Privacy Technique	Prosecutor	Journalist	Marketer	Singling Out	Rec. Linkage	Inference
<i>k</i> -Anonymity	Stops	Stops	Partial	Stops	Stops	Fails
<i>l</i> -Diversity	Stops	Stops	Stops	Stops	Stops	Partial
<i>t</i> -Closeness	Stops	Stops	Stops	Stops	Stops	Stops
Data Swapping	Partial	Partial	Partial	Partial	Stops	Partial
Global DP	Stops	Stops	Stops	Stops	Stops	Stops
Synthetic Data (DP)	Stops	Stops	Stops	Stops	Stops	Stops
Homomorphic Enc.	Stops	Stops	Stops	Stops	Stops	Stops
Federated Learning	Stops	Stops	Stops	Stops	Stops	Stops

Table 1: Effectiveness of PETs against standard privacy attacks (Part 1).

Privacy Technique	Attr. Disclose	Differencing	Membership	Model Inversion
<i>k</i> -Anonymity	Fails	Fails	Fails	Fails
<i>l</i> -Diversity	Stops	Fails	Fails	Fails
<i>t</i> -Closeness	Stops	Fails	Fails	Fails
Data Swapping	Partial	Fails	Fails	Fails
Global DP	Stops	Stops	Stops	Stops
Synthetic Data (DP)	Stops	Stops	Stops	Stops
Homomorphic Enc.	Stops	Fails	Fails	Fails
Federated Learning	Stops	Stops	Stops	Stops

Table 2: Effectiveness of PETs against machine learning and advanced attacks (Part 2).